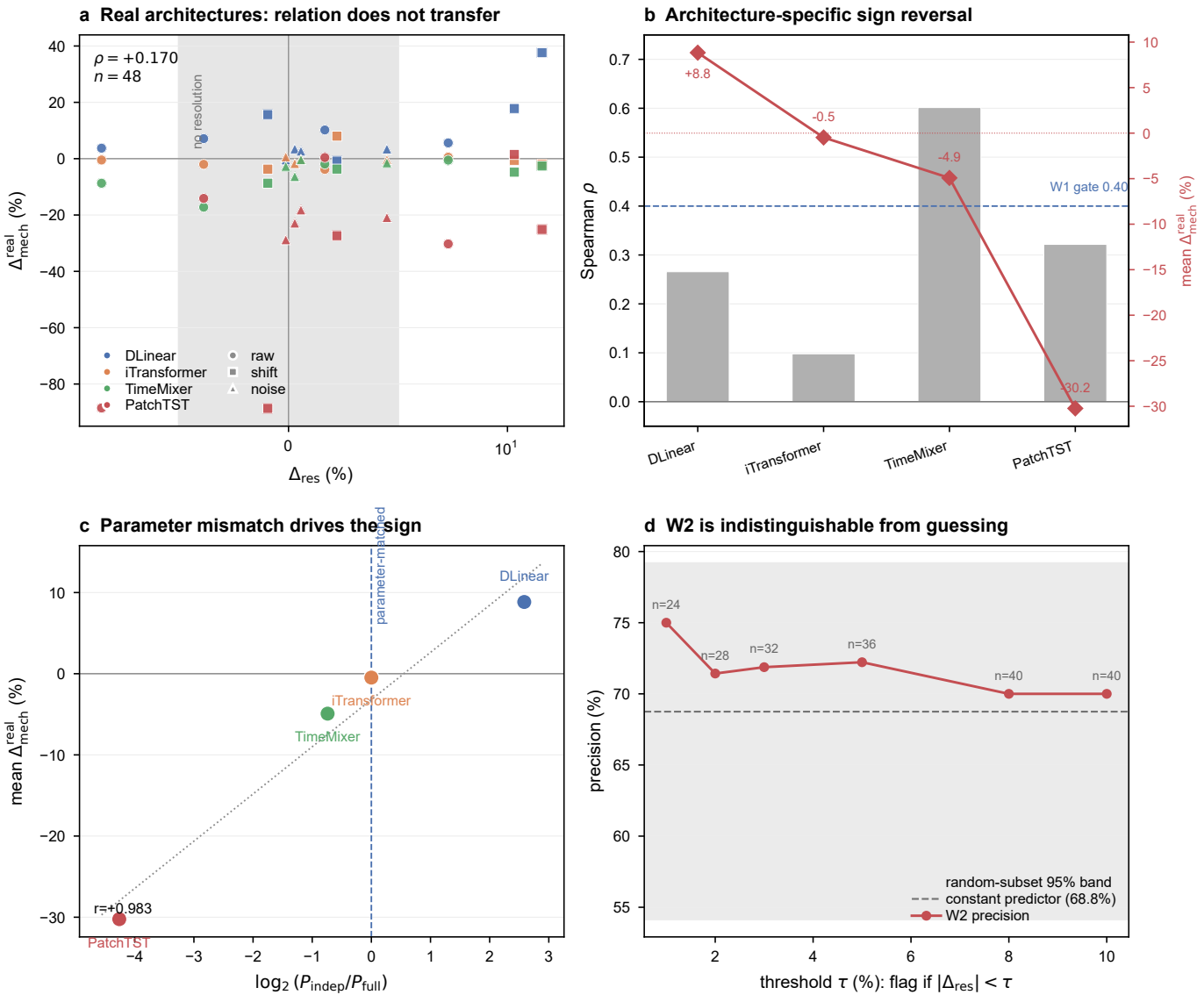

a Real architectures: relation does not transfer
ρ = +0.170
n = 48
no resolution
DLinear
iTransformer
TimeMixer
PatchTST
raw
shift
noise
Δ_mech^real (%)
Δ_res (%)
b Architecture-specific sign reversal
Spearman ρ
mean Δ_mech^real (%)
+8.8
-0.5
-4.9
-30.2
W1 gate 0.40
DLinear
iTransformer
TimeMixer
PatchTST
c Parameter mismatch drives the sign
parameter-matched
DLinear
iTransformer
TimeMixer
PatchTST
r=+0.983
mean Δ_mech^real (%)
log2 (P_indep/P_full)
d W2 is indistinguishable from guessing
precision (%)
n=24
n=28
n=32
n=36
n=40
n=40
random-subset 95% band
constant predictor (68.8%)
W2 precision
threshold τ (%): flag if |Δ_res| < τ